

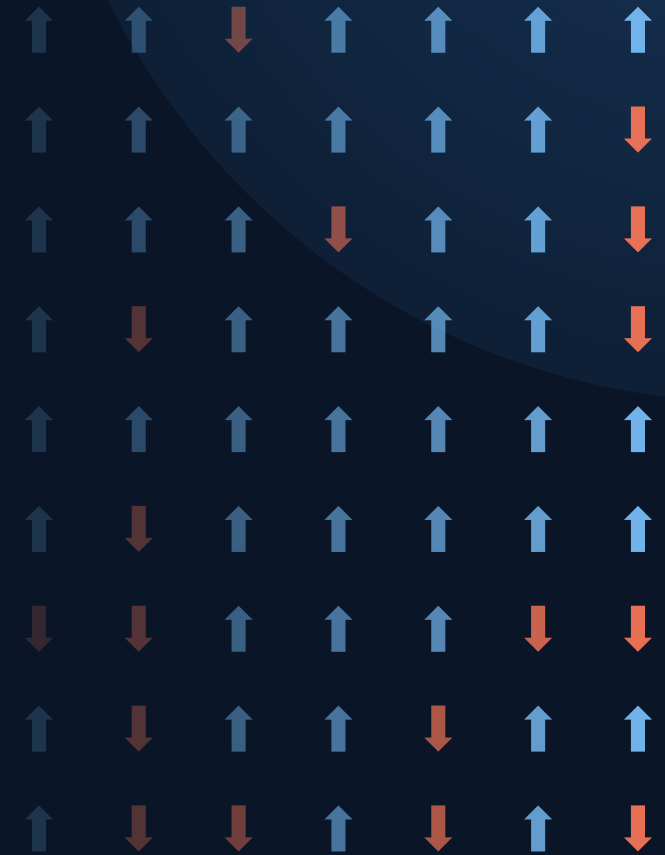
Measuring the External Field

A cortical-proxy content observable and the mean-field bound on
media-driven opinion change

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$$m = \tanh(\beta J m + h), \quad h = \alpha X$$



Media is modelled as a field on opinion. That field is assumed, never measured

WIRE REPORT

Federal Reserve holds interest rates steady, citing a stable inflation outlook.

informs

THE SAME EVENT

FED DESTROYS AMERICA. Your savings are GONE. The collapse they hid from you!

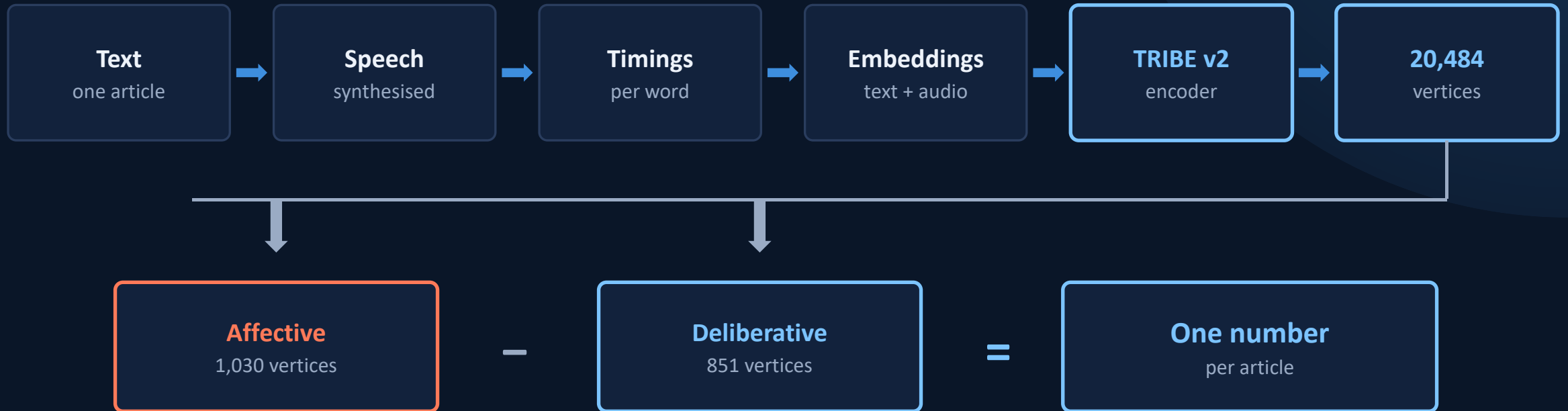
provokes

Readers notice the difference and cannot quantify it. Sentiment, readability and stance measure the words, not the response.

This project measures the field from content itself, applies it to 400 articles, and derives how strong the coupling must be for that spread to move a population at all.



A five-stage cascade turns one article into one number

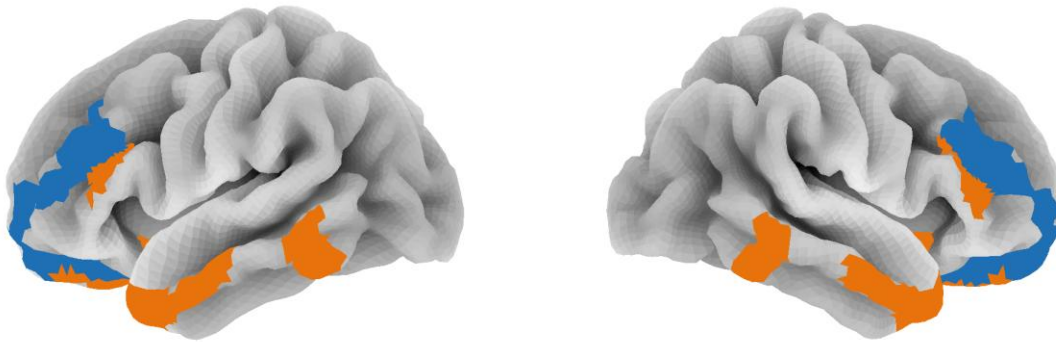


Encoder: TRIBE v2, d'Ascoli et al., Meta FAIR (2025), arXiv:2507.22229. Regions from HCP MMP1.0, Glasser et al. (2016). Instrument at monarch-4iy.pages.dev



The index contrasts two cortical networks, 1,030 vertices against 851

ROI definition, not a measurement: affective-salience (1030 vertices, orange) and deliberative-control (851, blue)
left right



THE OBSERVABLE

$$NAA_{\text{signed}} = A_{\text{aff}} - A_{\text{del}}$$

> 0 emotion leads

< 0 reasoning leads

A cortical proxy: it rates content, never a person, and is never called the amygdala. Regions resolved once and cached, so all 400 items average over identical vertices.

Regions on HCP MMP1.0 (Glasser et al., 2016); fsaverage5 surfaces.



Inspecting the checkpoint changed what the project could claim

01

Cortical only

No subcortical output.

No amygdala, so the proposal's equation cannot be computed.

02

Averaged over subjects

The loader forces it.

It predicts a typical viewer, which Paper 3 tests.

03

Standardised output

Means sit near zero, so a ratio breaks.

Undefined for 69 of 400 items; the difference replaces it.

All three were found by testing the checkpoint, before any result.



The corpus is length-matched and was powered before the scan

100

fear-activating

ISOT-fake

100

high outrage

SemEval-2019 T4

100

reward hook

Webis-Clickbait-17

100

neutral informational

PubMed + ISOT-true

≈ 164 words

length-matched

means 167.4, 163.5, 163.7, 162.2; sd near 11

$\eta^2 \geq 0.0268$

detectable, fixed before the scan

smallest detectable AUC = 0.5916

Corpora: Ahmed et al. (ISOT, 2017); Kiesel et al. (SemEval-2019 Task 4); Potthast et al. (Webis-Clickbait-17); PubMed open abstracts.

THE FOUR CATEGORIES SEPARATE AT

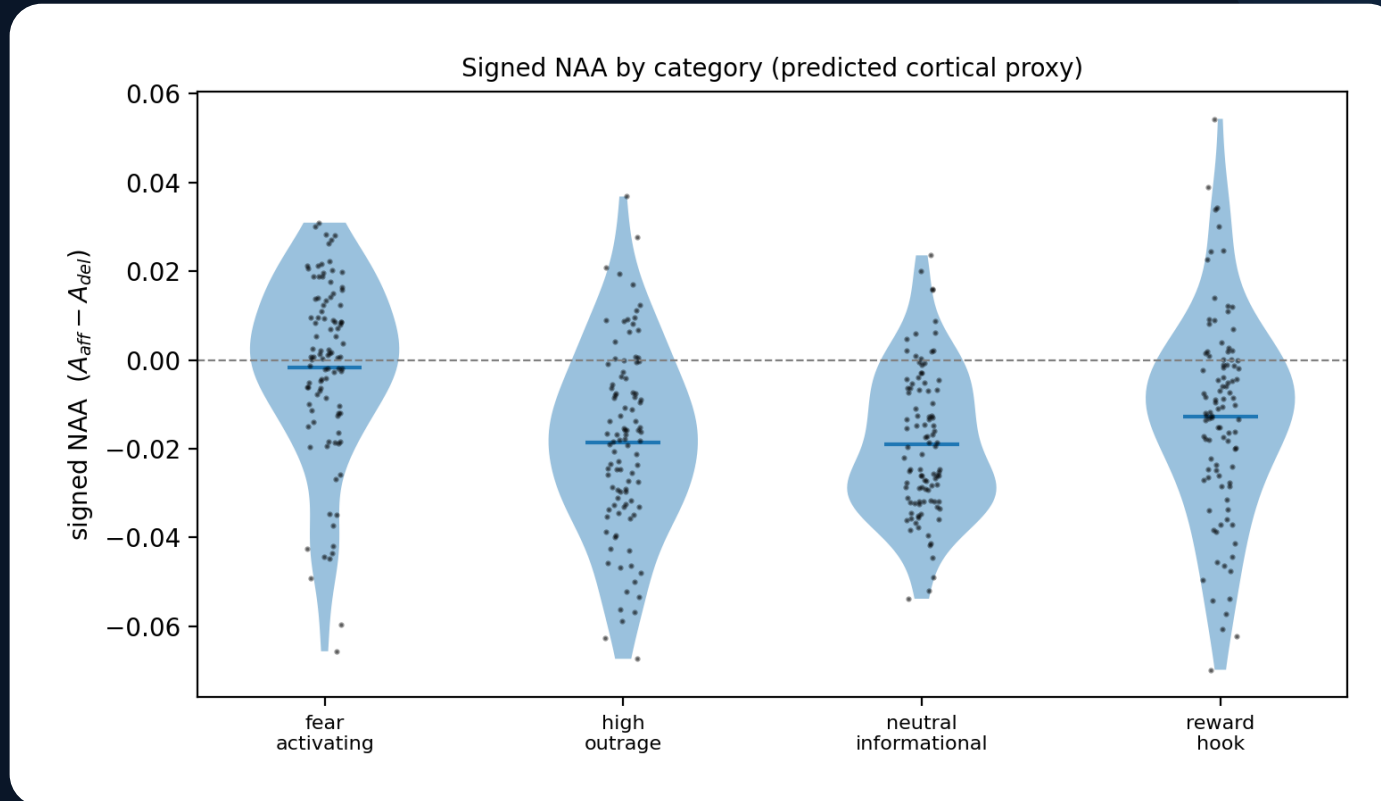
$$\eta^2 = 0.1068$$

$F = 15.779$ $p = 1.03 \times 10^{-9}$ $n = 400$





Categories shift against each other, but the distributions overlap

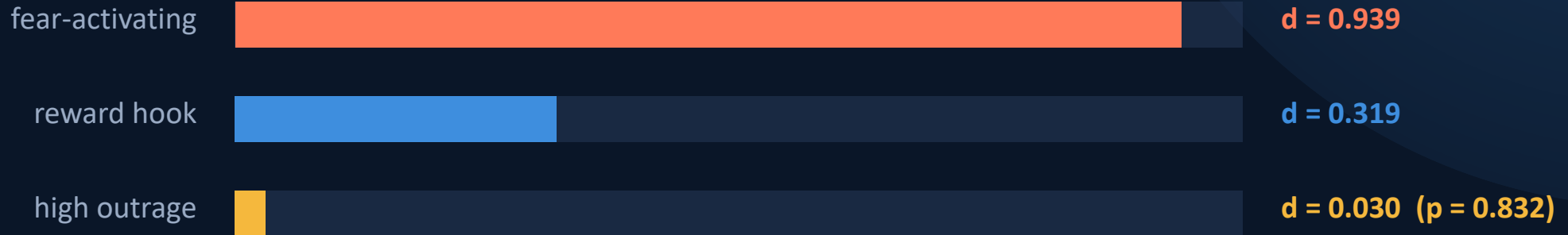


Every category mean is negative: the deliberative network leads throughout, and the categories differ in how far below zero they sit.



Fear-activating content carries the effect; outrage moves both networks equally

COHEN'S d AGAINST THE NEUTRAL BASELINE



OUTRAGE, TAKEN APART

Affective $d = +0.507$, deliberative $d = +0.491$. A flat index here is a symmetric response, not an absent one. Fear raises one network only ($d = +0.613$ and -0.098).

⚠ Category is confounded with source, and a bag of words beats the index

CATEGORY		SOURCE DATASET
fear-activating	→	ISOT-fake
high outrage	→	SemEval-2019 T4
reward hook	→	Webis-Clickbait-17
neutral	→	PubMed + ISOT-true

0.6274

AUC, the index
nothing fitted

0.9758

AUC, TF-IDF
logistic, 5-fold

Same 400 rows, same label. VADER sits at chance, 0.5392. The same words guess the source 66% of the time against 25%, so much of the lexical win is provenance.

Length is controlled; provenance is not. **The index is a field observable for the physics, not a detector.**

Hutto & Gilbert (2014), VADER. One source per category, so the separation cannot be attributed to framing.



Group-level claims hold at this precision; per-item claims do not

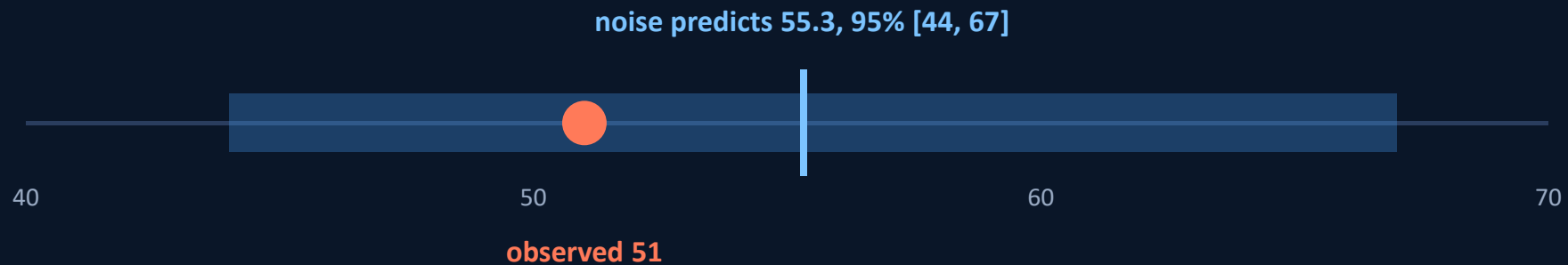
0.8725

ICC between two GPU sessions
separation replicates: $\eta^2 = 0.0888$ against 0.1068

12.8%

of items reverse direction
51 of 400; no verdict on one article is ever made

REVERSALS: OBSERVED AGAINST WHAT MEASURED NOISE ALONE PREDICTS





Media enters as a field $h = \alpha X$, and the corpus cannot pin down α

THE MODEL

Opinions ± 1 , pulled toward neighbours by J , pushed by media with field $h = \alpha X$.

$$m = \tanh(\beta J m + h)$$

$$F(m) = a m^2 + b m^4 - h m$$

$$a = (1 - \beta J)/2, \quad b = 1/12$$

Coefficients derived, correcting the proposal's a and b .

THE COUPLING

Tried. Fit the outcome to the index and read off α .

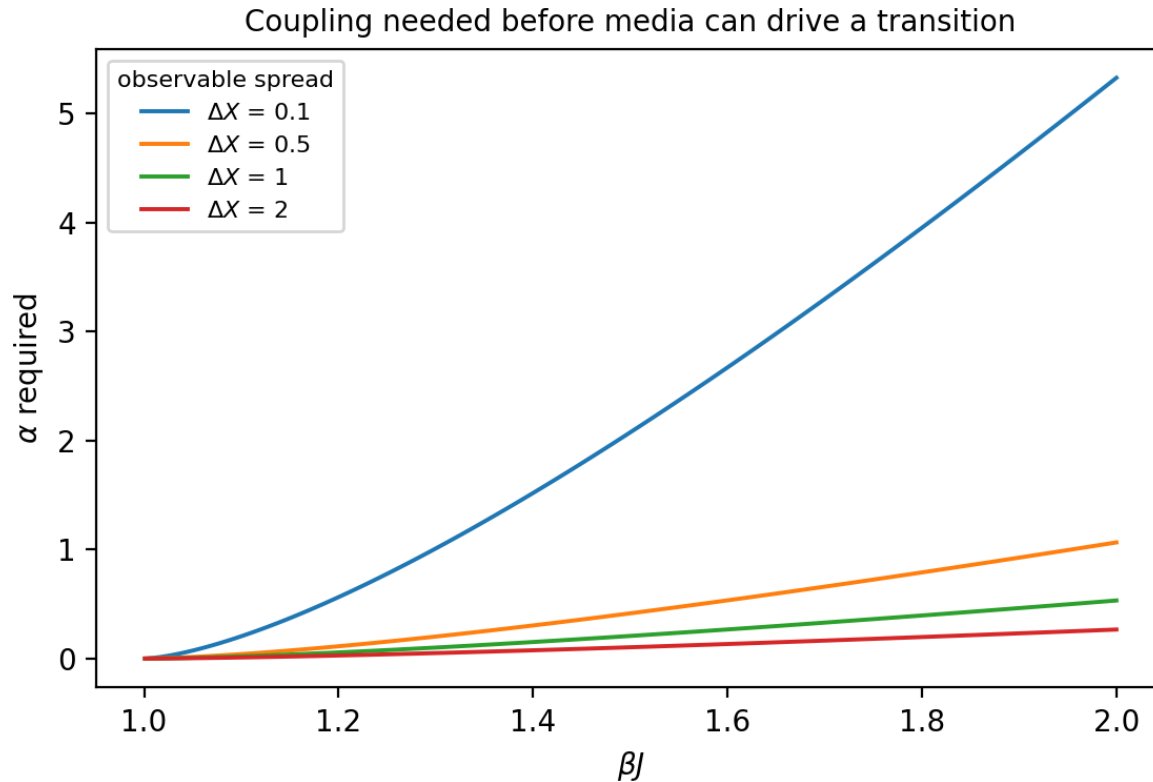
Result. Out of sample the fit does worse than the average, and the estimate depends on βJ , which nothing here measures.

So no value of α is quoted anywhere.

What replaces it needs no fit at all: the bound.



Any content observable must clear a bound to drive a transition



$$\alpha \geq h_c(\beta J) / \Delta X$$

What α would have to be for media to tip opinion, for any observable, before any data. Within the mean-field model.

$$\alpha \geq 4.29$$

at $\beta J = 2$

measured $\Delta X = 0.1241$

✓ Four claims the work supports, and four it does not

✓ SUPPORTED

- A content observable that can be measured, with its reliability stated.
- That it separates these four corpora at 0.1068, power attached.
- A bound any candidate observable must satisfy, usable before data exists.
- A noise ceiling on public fMRI (0.1517), and a first check of the encoder against it.

✗ NOT SUPPORTED

- That the index detects manipulation. Source is confounded; TF-IDF scores 0.9758.
- Anything about the amygdala. The checkpoint does not predict it.
- Any value of the coupling α . The corpus does not identify it.
- Any verdict on one article. 12.8% reverse between sessions.

Conclusions

- 1. A field observable can be measured from content.** The instrument runs, the corpus is complete, and reliability is measured: ICC = 0.8725.
- 2. It separates four corpora at 0.1068, confounded with source.** Both halves are reported, the confound first.
- 3. The coupling is unidentified, so the thesis reports a bound.** With $\Delta X = 0.1241$, $\alpha \geq 4.294$ at $\beta J = 2$: a constraint on any future proposal of this mechanism.
- 4. The encoder is weakly positive against real brains.** $r = +0.028$ against a ceiling of 0.096 on the same scans; not yet decisive.

81 pp Dissertation

12 pp Paper 1 · Physica A

16 pp Paper 2 · Physica A

8 pp Paper 3 · Imaging
Neurosci.

Every number comes from a re-runnable script · github.com/brn-mwai/monarch · monarch-4iy.pages.dev